COMMUNICATION-CONSTRAINED MULTI-AGENT EXPLORATION

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ABSTRACT

We study cooperative exploration by multiple agents in unknown grid environments when communication resources are limited. Each agent observes only a local portion of the environment and must coordinate with other agents to achieve high coverage while reducing redundant exploration. We formulate the task as a Decentralised Partially Observable Markov Decision Process with Communication and study the effect of a normalised communication budget on exploration performance. Our proposed approach combines Multi-Agent Proximal Policy Optimisation (MAPPO) with a differentiable discrete communication bottleneck and a budget-aware communication scheduler. The planned experiments evaluate coverage, time-to-95%-coverage, overlap, success rate, bandwidth usage, and communication efficiency across different communication budgets and environment settings.

***K*eywords** Multi-Agent Reinforcement Learning · MAPPO · Multi-Agent Exploration · Communication Constraints · Decentralised POMDP

1 Introduction

Multi-agent exploration enables a team of autonomous agents to explore unknown environments collaboratively, with applications such as search-and-rescue, drone swarms, and autonomous mapping. Since each agent has only a local view of the environment, communication is important for sharing information and avoiding redundant exploration. However, communication may be constrained by limited bandwidth, range, energy, or frequency, creating a trade-off between communication cost and exploration performance.

We formulate the task as a Decentralised Partially Observable Markov Decision Process with Communication (Dec-POMDP-Comm) [3] and use Multi-Agent Proximal Policy Optimisation (MAPPO) with decentralised actors and a centralised critic [1]. A differentiable discrete communication bottleneck [4] and a budget-aware communication scheduler are used to model constrained communication. The project investigates how reducing communication resources affects cooperative exploration efficiency.

The main research question is:

> *How does a constrained communication budget affect the efficiency of cooperative multi-agent exploration?*

2 Preliminaries

2.1 Dec-POMDP with Communication

We model the multi-agent exploration task as

$$M = \left\langle S, \{A_i\}_{i=1}^{N}, \{O_i\}_{i=1}^{N}, \{M_i\}_{i=1}^{N}, P, O, R, \gamma, \beta \right\rangle, \tag{1}$$

where S is the global state space, A_i is the action space of agent i, O_i is its local observation space, and M_i is its discrete message space. P denotes the transition model, O the observation model, R the shared reward, γ the discount factor, and β the available communication budget.

Symbol	Definition
S	Global state space
A_i	Action space of agent i
O_i	Local observation of agent i
M_i	Discrete message space of agent i
P	State transition probability
O	Observation probability
R	Shared reward function
γ	Discount factor
β	Normalised communication budget

Table 1: Notation used for the Dec-POMDP with communication.

At timestep t, agent i receives an observation o_i^t, chooses an action a_i^t, and can transmit a discrete message m_i^t. The agents share the same exploration objective.

2.2 Communication Budget

The communication budget is represented by a normalised quantity

$$\beta = \frac{d_m}{d_m^{\max}} \frac{f_c}{f_c^{\max}} \frac{|E|}{|E|^{\max}}, \tag{2}$$

where d_m is the message dimension, f_c is the communication frequency, and $|E|$ is the number of communicating agent pairs. The maximum number of possible undirected communication pairs is

$$|E|^{\max} = \frac{N(N-1)}{2}. \tag{3}$$

The message dimensions considered are

$$d_m \in \{1, 2, 4, 8, 16, 32\}, \tag{4}$$

and the experiments use the budget levels

$$\beta \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}. \tag{5}$$

2.3 Exploration Metrics

Let C_t be the set of grid cells covered by at least one agent by time t, and let G denote the set of traversable cells. Coverage is

$$\mathrm{Cov}(t) = \frac{|C_t|}{|G|}. \tag{6}$$

The time-to-95%-coverage is defined as

$$T_{95} = \min\{t : \mathrm{Cov}(t) \geq 0.95\}. \tag{7}$$

Overlap measures redundant exploration caused by agents visiting regions that have already been explored by other agents.

3 Problem Formulation

Consider N homogeneous agents exploring an unknown $G \times G$ grid. Each agent has a local $k \times k$ field of view.

The global state is

$$s_t = [\mathrm{map}, \{p_i^t\}_{i=1}^N, \{\mathrm{visited}_i^t\}_{i=1}^N], \tag{8}$$

where p_i^t denotes the position of agent i.

The local observation of agent i is

$$o_i^t = [\text{local map}, p_i^t, \text{visited mask}, \text{neighbour information}]. \tag{9}$$

The action space is

$$a_i^t \in \{\mathrm{up}, \mathrm{down}, \mathrm{left}, \mathrm{right}\}. \tag{10}$$

The message transmitted by agent i is

$$m_i^t \in M_i, \tag{11}$$

subject to the communication constraint

$$C_{\text{comm}} \leq C_{\text{budget}}(\beta). \tag{12}$$

The joint action is

$$\mathbf{a}^t = (a_1^t, \ldots, a_N^t), \tag{13}$$

and the joint policy is

$$\boldsymbol{\pi}_\theta = (\pi_{\theta_1}, \ldots, \pi_{\theta_N}). \tag{14}$$

The optimisation objective is

$$\max_\theta \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R_t\right]. \tag{15}$$

The shared reward is defined as

$$R_t = w_c \Delta\text{Cov}_t - w_o \text{Overlap}_t - w_t - w_m C_t^{\text{comm}} + B_{\text{success}}, \tag{16}$$

where ΔCov_t is the coverage increase, Overlap_t is redundant exploration, C_t^{comm} is communication cost, and B_{success} is a success bonus when final coverage satisfies

$$\text{Cov}(T) \geq 0.95. \tag{17}$$

The centralised critic receives the global state

$$V_\phi(s_t), \tag{18}$$

while each actor uses only its local observation

$$\pi_\theta(a_i^t | o_i^t). \tag{19}$$

Thus, the learning problem is to obtain a joint policy that maximises exploration performance while satisfying a constrained communication budget.

4 Proposed Solutions

4.1 MAPPO

We use Multi-Agent Proximal Policy Optimisation (MAPPO) with shared actor parameters and a centralised critic during training [1]. Since the agents are homogeneous, parameter sharing allows the same policy to be used by all agents.

The PPO probability ratio is

$$r_t(\theta) = \frac{\pi_\theta(a_t|o_t)}{\pi_{\theta_{\text{old}}}(a_t|o_t)}. \tag{20}$$

The clipped policy objective is

$$L^{\text{CLIP}} = \mathbb{E}_t\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{21}$$

where $\hat{A}_t$ is the estimated advantage and ϵ is the PPO clipping parameter [2].

The critic is trained using the global state while the actors operate using local observations.

4.2 Differentiable Communication Bottleneck

For each agent, the local observation is encoded as

$$z_i^t = f_{\text{enc}}(o_i^t). \tag{22}$$

The latent representation is converted into a discrete message through a quantisation operation:

$$m_i^t = Q(z_i^t). \tag{23}$$

The discrete message is transmitted to neighbouring agents. A straight-through estimator is used to allow gradient-based optimisation through the discrete communication operation during training. This design is based on differentiable discrete communication bottlenecks [4].

The combined optimisation objective is

$$L_{\text{total}} = L_{\text{MAPPO}} + \lambda_{\text{comm}} L_{\text{comm}}, \tag{24}$$

where L_{comm} represents the communication objective and λ_{comm} controls its contribution.

4.3 Budget-Constrained Communication

Each agent estimates the usefulness of communicating at timestep t using

$$u_i^t = \sigma(W_u h_i^t + b_u), \tag{25}$$

where h_i^t is the internal agent representation and σ is the sigmoid function.

A message is transmitted when

$$u_i^t > \tau, \tag{26}$$

where τ is determined according to the target communication budget. This allows the policy to learn when information exchange is useful rather than communicating at every timestep. Learning communication decisions in discrete communication channels has been studied in prior work [5].

4.4 Environment

The experimental environment consists of procedurally generated grid worlds with dimensions between 20×20 and 50×50 cells. Obstacle density is varied between 10% and 30%.

The number of agents is

$$N \in \{3, 4, 5\}. \tag{27}$$

The local field of view is varied between 5×5 and 7×7. Agents communicate with neighbouring agents within a Manhattan distance of up to three cells.

The main training configuration uses:

- Actor learning rate: 3×10^{-4}
- Critic learning rate: 10^{-3}
- Discount factor: $\gamma = 0.99$
- GAE parameter: $\lambda = 0.95$
- PPO clipping parameter: $\epsilon = 0.2$
- Entropy coefficient: 0.01
- Hidden representation: 64-unit MLP / 128-unit GRU
- Training: multiple random seeds

4.5 Experimental Plan

The main experiments are designed to measure how communication constraints affect exploration.

Experiment	$\boldsymbol{\beta}$	**Agents**	**Purpose**
Budget sweep	$\{0, 0.1, 0.25, 0.5, 0.75, 1\}$	4	Communication trade-off
Agent scaling	0.25	3,4,5	Team-size effect
Grid scaling	0.25	4	Environment complexity
Communication range	0.5	4	Range effect
Ablation	0.25	4	Communication mechanism

Table 2: Planned experimental settings.

The evaluation metrics are

- T_{95}: time required to reach 95% coverage.
- Coverage: fraction of traversable cells explored.
- Overlap: redundant exploration between agents.
- Success rate: fraction of runs reaching 95% coverage.
- Actual bandwidth: communication transmitted during exploration.
- Exploration rate: coverage obtained per unit time.
- Communication efficiency: coverage gain per transmitted bit.

For each communication budget, the measured bandwidth and exploration performance will be used to construct a bandwidth–performance Pareto frontier. The knee of the frontier will be identified using the Kneedle method [6].

The learned communication protocol will also be analysed using message frequency, message entropy, and mutual information between transmitted messages and relevant state features. These analyses are intended to characterise what information agents learn to communicate when bandwidth is limited.

5 Contribution

Member	**Responsibility**	**Major Tasks**
Govindu Srimaan	MAPPO and Communication	MAPPO, communication module, quantisation, budget mechanism
Maru Jainil	Environment and Reward	GridWorld, observations, coverage, overlap, reward
Manavi	Training and Experiments	Training pipeline, β sweep, experiments, logging
Priyanka	Evaluation and Analysis	Metrics, Pareto/knee analysis, protocol analysis, plots

Table 3: Division of project responsibilities.

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